

IBC Short Course

Module 2: Individualized Treatment Rules

Single, Multiple, and Indefinite-Horizon Regimes

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Outline

1. **Single-Decision Setting**

rules, value, identification, and estimation

2. **Multi-Decision Setting**

extending from a single timepoint, backward induction

3. **Infinite Time Horizon**

MDPs, reinforcement learning

*“... the modern concept of precision medicine, the paradigm wherein patient heterogeneity is leveraged through data-driven approaches to improve treatment decisions so that the **right treatment** is given to the **right patient** at the **right time**.”*

— Kosorok & Laber (2019)

From HTE to Treatment Policies

Heterogeneous
treatment effects

What happened?
For whom did
treatment help?

Individualized
treatment rules

What should we do?
Which action
is best now?

A DTR turns evidence of variation (such as the HTE/ATE)
into treatment actions and learns how to evaluate them.

Thresholding of the CATE is different than recommending actions.

Section 1

Single-Decision Setting

Single Decision Notation

- ▶ Observed data: (H_{1i}, A_{1i}, Y_i) , $i = 1, \dots, n$.
- ▶ $H_1 = X_1$: history available at the single decision point.
- ▶ $A_1 \in \mathcal{A}_1$: treatment or action actually received.
- ▶ Y : outcome, coded so larger values are better.

A single-decision regime is $d = \{d_1(h_1)\}$.

The rule $d_1 : \mathcal{H}_1 \rightarrow \mathcal{A}_1$ maps history to an action.

Static vs Dynamic Regimes

Static regime

$$d_1(h_1) \equiv 1$$

$$d_1(h_1) \equiv 0$$

Same action for everyone.

Dynamic regime

$$d_1(h_1) = \mathbb{I}(\text{age} < 50, \text{WBC} < 10)$$

$$d_1(h_1) = \mathbb{I}\{s_\eta(h_1) > 0\}$$

Action depends on available history.

“Dynamic” refers to leveraging the heterogeneity in patient histories, even when there’s only one decision.

“Dynamic” used to refer to multiple decisions as opposed to a single one.

Potential Outcomes Under a Rule

For two actions, each individual has potential outcomes $Y^*(1)$ and $Y^*(0)$.

If a rule d were used, the potential outcome would be

$$\begin{aligned} Y^*(d) &= Y^*(1)\mathbb{I}\{d_1(H_1) = 1\} + Y^*(0)\mathbb{I}\{d_1(H_1) = 0\} \\ &= Y^*(1)d_1(H_1) + Y^*(0)\{1 - d_1(H_1)\}. \end{aligned}$$

The rule chooses which potential outcome is relevant for someone with history H_1 .

Value of a Rule

The **value** of a rule/policy/decision/DTR is the mean potential outcome if the rule were applied:

$$V(d) = \mathbb{E}\{Y^*(d)\}.$$

An optimal rule in a class $\mathcal{D}$ satisfies

$$d^{\text{opt}} \in \operatorname{argmax}_{d \in \mathcal{D}} V(d).$$

Evaluation vs. Learning

Policy evaluation

given d , $V(d) = \mathbb{E}\{Y^*(d)\}$.

Estimate how well a specified rule would perform.

Policy learning

$$d_{\eta}^{\text{opt}} \in \operatorname{argmax}_{d_{\eta} \in \mathcal{D}_{\eta}} V(d_{\eta}).$$

Estimate a rule with high value in a chosen class $\mathcal{D}_{\eta}$.

What is the class $\mathcal{D}_{\eta}$? A family of candidate rules indexed by parameters η . For $X = (X_1, X_2)^{\top}$, for example:

hyperplanes: $d_{\eta}(X) = \mathbb{I}(\eta_{11} + \eta_{12}X_1 + \eta_{13}X_2 > 0)$,

rectangular regions: $d_{\eta}(X) = \mathbb{I}(X_1 < \eta_{11}) + \mathbb{I}(X_1 \geq \eta_{11}, X_2 < \eta_{12})$.

The class fixes the *shape* of rule we can learn — and its interpretability; learning just searches over the parameters η .

Observed Data and the Counterfactual Target

We observe independent copies of

$$O_i = (X_{1i}, A_{1i}, Y_i), \quad H_{1i} = X_{1i}, \quad i = 1, \dots, n.$$

For a fixed regime $d = \{d_1(h_1)\}$, the causal target is

$$V(d) = \mathbb{E}\{Y^*(d)\}.$$

How to identify: Can this counterfactual mean be written using the distribution of (H_1, A_1, Y) ?

Potential Outcomes Under a Fixed Rule

For a finite action set $\mathcal{A}_1$, define

$$Y^*(d) = \sum_{a_1 \in \mathcal{A}_1} Y^*(a_1) \mathbb{I}\{d_1(H_1) = a_1\}.$$

For $\mathcal{A}_1 = \{0, 1\}$,

$$Y^*(d) = Y^*(1)d_1(H_1) + Y^*(0)\{1 - d_1(H_1)\}.$$

The regime chooses which potential outcome is relevant for each realized history.

What We Can Observe

The observed outcome is the potential outcome under the action actually received:

$$Y = \sum_{a_1 \in \mathcal{A}_1} Y^*(a_1) \mathbb{I}(A_1 = a_1).$$

For a fixed rule d , define consistency with the rule:

$$C_d = \mathbb{I}\{A_1 = d_1(H_1)\}.$$

If $C_d = 1$, then the observed outcome agrees with the outcome under rule d . If $C_d = 0$, that counterfactual outcome is not directly observed.

Assumptions

- ▶ **SUTVA / consistency:** observed outcomes equal the relevant potential outcomes, with no interference or hidden treatment versions.
- ▶ **No unmeasured confounders:**
 $\{Y^*(a_1) : a_1 \in \mathcal{A}_1\} \perp\!\!\!\perp A_1 \mid H_1.$
- ▶ **Positivity:** $P(A_1 = a_1 \mid H_1 = h_1) > 0$ for treatment options that must be learned at histories with support.

Together, these let us replace potential outcomes with things we can observe in the data.

Assumption 1: SUTVA / Consistency

For two actions, consistency can be written as

$$Y_i = Y_i^*(1)A_{1i} + Y_i^*(0)(1 - A_{1i}).$$

- ▶ If $A_{1i} = 1$, then $Y_i = Y_i^*(1)$.
- ▶ If $A_{1i} = 0$, then $Y_i = Y_i^*(0)$.
- ▶ No individual's outcome is affected by another individual's assigned action.
- ▶ The action labels correspond to well-defined treatment options.

Assumption 2: No Unmeasured Confounders

The single-decision assumption is

$$\{Y^*(1), Y^*(0)\} \perp\!\!\!\perp A_1 \mid H_1.$$

- ▶ Within levels of history, treatment assignment is as good as random.
- ▶ H_1 must include common causes of treatment assignment and outcome.
- ▶ **No interference:** one individual's outcome is unaffected by others' assigned actions — so $Y^*(a_1)$ depends only on that individual's own action.
- ▶ In a randomized study, this is built into the design.
- ▶ In observational data, this is not testable from the observed data alone.

Assumption 3: Positivity

For histories that occur in the target population,

$$P(A_1 = a_1 \mid H_1 = h_1) > 0, \quad a_1 \in \mathcal{A}_1.$$

- ▶ To evaluate a fixed rule, the action $d_1(h_1)$ must be possible at that history.
- ▶ To learn an unrestricted optimal rule, all candidate actions must be possible.
- ▶ Positivity can hold in theory, but small proportions of treatments can lead to unstable estimation.

From Assumptions to an Estimable Target

From our assumptions, the goal is to rewrite the counterfactual value $V(d) = \mathbb{E}\{Y^*(d)\}$ using only the observed data.

We can estimate the causal effects three ways: with regression (Q-learning) and with weighting (IPW), or both (AIPW).

Define the observed-data regression “Q-function”

$$Q_1(h_1, a_1) = \mathbb{E}(Y \mid H_1 = h_1, A_1 = a_1).$$

$$\begin{aligned} V(d) &= \mathbb{E}\{Y^*(d)\} \\ &= \mathbb{E}\left[\sum_{a_1 \in \mathcal{A}_1} \mathbb{E}\{Y^*(a_1) \mid H_1\} \mathbb{I}\{d_1(H_1) = a_1\}\right] \\ &= \mathbb{E}\left[\sum_{a_1 \in \mathcal{A}_1} Q_1(H_1, a_1) \mathbb{I}\{d_1(H_1) = a_1\}\right]. \end{aligned}$$

Identification via Outcome Regression

Once Q_1 identifies the conditional mean potential outcomes,

$$d_1^{\text{opt}}(h_1) \in \operatorname{argmax}_{a_1 \in \mathcal{A}_1} Q_1(h_1, a_1).$$

For $\mathcal{A}_1 = \{0, 1\}$ with option 0 as the default,

$$d_1^{\text{opt}}(h_1) = \mathbb{I}\{Q_1(h_1, 1) > Q_1(h_1, 0)\}.$$

The maximum value is

$$V(d^{\text{opt}}) = \mathbb{E} \left\{ \max_{a_1 \in \mathcal{A}_1} Q_1(H_1, a_1) \right\}.$$

Q_1 is now our estimation target

Estimating the Q -Function

A working model $Q_1(h_1, a_1; \beta_1)$ is fit to the observed data.
For continuous outcomes, ordinary least squares solves

$$\sum_{i=1}^n \frac{\partial Q_1(H_{1i}, A_{1i}; \beta_1)}{\partial \beta_1} \{Y_i - Q_1(H_{1i}, A_{1i}; \beta_1)\} = 0.$$

After estimating $\hat{\beta}_1$, choose the action with the larger fitted mean outcome.

Binary Linear Q-Learning

For $\mathcal{A}_1 = \{0, 1\}$, a common working model is

$$Q_1(h_1, a_1; \beta_1) = \beta_{11} + \beta_{12}^T h_1 + \beta_{13} a_1 + \beta_{14}^T h_1 a_1.$$

Then

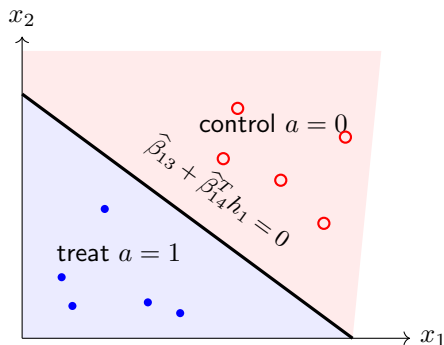
$$Q_1(h_1, 1; \beta_1) - Q_1(h_1, 0; \beta_1) = \beta_{13} + \beta_{14}^T h_1,$$

so the estimated rule is

$$\hat{d}_{Q,1}^{\text{opt}}(h_1) = \mathbb{I}\{\hat{\beta}_{13} + \hat{\beta}_{14}^T h_1 > 0\}.$$

The Induced Decision Boundary

A linear Q -model gives a linear rule $\hat{d}_1(h_1) = \mathbb{I}\{\hat{\beta}_{13} + \hat{\beta}_{14}^T h_1 > 0\}$.



The model class fixes the boundary's shape — linear here, more flexible for trees or forests.

Q-Learning Algorithm: Single Decision

1. Specify a working model $Q_1(h_1, a_1; \beta_1)$.
2. Estimate $\hat{\beta}_1$ from (H_{1i}, A_{1i}, Y_i) .
3. For a new history h_1 , compute fitted values for each action:

$$Q_1(h_1, 0; \hat{\beta}_1), \quad Q_1(h_1, 1; \hat{\beta}_1).$$

4. Recommend the action with the larger fitted value.

Estimating the Value After Q-Learning

The fitted optimal value can be estimated by plug-in:

$$\hat{V}_Q(\hat{d}_Q^{\text{opt}}) = \frac{1}{n} \sum_{i=1}^n \max_{a_1 \in \mathcal{A}_1} Q_1(H_{1i}, a_1; \hat{\beta}_1).$$

Alternatively, evaluate the learned rule using IPW or AIPW:

$$\hat{V}_{\text{AIPW}}(\hat{d}_Q^{\text{opt}}).$$

What Class of Rules Did We Learn?

The working model implies a rule class:

$$\hat{d}_{Q,1}^{\text{opt}}(h_1) = \operatorname{argmax}_{a_1} Q_1(h_1, a_1; \hat{\beta}_1).$$

- ▶ Linear interactions induce linear decision boundaries.
- ▶ Tree or forest models induce more flexible boundaries.
- ▶ Misspecifying outcome levels can still distort the decision boundary.

Q-Learning: Practical Considerations

- ▶ Straightforward to implement with standard regression tools.
- ▶ Efficient if the outcome regression model is correctly specified.
- ▶ The fitted model induces the class of rules under consideration.
- ▶ Misspecifying Q_1 can produce a poor estimated rule, even if the induced class is reasonable.
- ▶ The max operation creates nonregular inference.

Optimal Decision $\neq$ Best Decision

$d_1^{\text{opt}}(H_1)$ chooses the action with best expected outcome given H_1 .

It does not reveal the best realized potential outcome for each individual:

$$Y^*(d^{\text{opt}}) \leq \max\{Y^*(1), Y^*(0)\}.$$

Optimality is relative to the information available at the decision time.

Identification via Weighting

For a fixed binary-action rule d , let

$$\begin{aligned}C_d &= \mathbb{I}\{A_1 = d_1(H_1)\}, \\ \pi_1(h_1) &= P(A_1 = 1 \mid H_1 = h_1), \\ \pi_{d,1}(H_1) &= P(C_d = 1 \mid H_1).\end{aligned}$$

Then

$$\pi_{d,1}(H_1) = \pi_1(H_1)\mathbb{I}\{d_1(H_1) = 1\} + \{1 - \pi_1(H_1)\}\mathbb{I}\{d_1(H_1) = 0\}.$$

Under the same assumptions,

$$V(d) = \mathbb{E} \left\{ \frac{C_d Y}{\pi_{d,1}(H_1)} \right\}.$$

Reweighting subjects by their chance of following the rule is **inverse probability weighting (IPW)**.

IPW Estimator

In data, replace the population mean by an empirical mean:

$$\hat{V}_{\text{IPW}}(d) = \frac{1}{n} \sum_{i=1}^n \frac{C_{d,i} Y_i}{\pi_{d,1}(H_{1i}; \hat{\gamma}_1)}.$$

- ▶ In randomized data, $\pi_1(h_1)$ is known by design.
- ▶ In observational data, we estimate $\pi_1(h_1; \gamma_1)$.
- ▶ Correct propensity modeling is the key modeling requirement.

People who happened to receive the action recommended by the rule can represent everyone, after weighting by the chance of consistency with the rule.

▶ Derivation: why $\hat{V}_{\text{IPW}}$ targets $V(d)$

Normalized IPW

A common finite-sample modification is the normalized weighted average:

$$\hat{V}_{\text{IPW}^*}(d) = \frac{\sum_{i=1}^n C_{d,i} Y_i / \pi_{d,1}(H_{1i}; \hat{\gamma}_1)}{\sum_{i=1}^n C_{d,i} / \pi_{d,1}(H_{1i}; \hat{\gamma}_1)}.$$

- ▶ Same target under correct propensity modeling.
- ▶ Often less variable in finite samples.
- ▶ Useful when the raw IPW weights are uneven.

IPW Diagnostics

- ▶ How many subjects have $C_d = 1$?
- ▶ How small are the estimated values of $\pi_{d,1}(H_{1i})$?
- ▶ Are the largest weights attached to unusual histories?
- ▶ Does the rule require treatment options rarely observed in parts of $\mathcal{H}_1$?

A simple effective-sample-size summary is

$$\frac{(\sum_i W_i)^2}{\sum_i W_i^2}, \quad W_i = \frac{C_{d,i}}{\widehat{\pi}_{d,i}}.$$

AIPW: Add an Outcome Model

Augmented IPW (AIPW) pairs an outcome model with the IPW weights.

Recall the outcome regression

$$Q_1(h_1, a_1) = \mathbb{E}(Y \mid H_1 = h_1, A_1 = a_1).$$

For a fixed rule d , define the rule-specific mean

$$\begin{aligned} m_d(H_1) &= Q_1(H_1, 1)\mathbb{I}\{d_1(H_1) = 1\} \\ &\quad + Q_1(H_1, 0)\mathbb{I}\{d_1(H_1) = 0\} \\ &= \mathbb{E}\{Y^*(d) \mid H_1\}. \end{aligned}$$

AIPW combines:

- ▶ prediction for everyone through $\hat{m}_d(H_1)$;
- ▶ residual correction among people with $C_d = 1$.

AIPW Estimator

Let

$$\hat{m}_d(H_{1i}) = Q_1(H_{1i}, d_1(H_{1i}); \hat{\beta}_1), \quad \hat{\pi}_{d,i} = \pi_{d,1}(H_{1i}; \hat{\gamma}_1).$$

The AIPW estimator can be written compactly as

$$\hat{V}_{\text{AIPW}}(d) = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_d(H_{1i}) + \frac{C_{d,i}}{\hat{\pi}_{d,i}} \{Y_i - \hat{m}_d(H_{1i})\} \right].$$

Prediction plus a weighted correction for prediction error.

AIPW: Equivalent Augmentation Form

The same estimator can be written as

$$\hat{V}_{\text{AIPW}}(d) = \frac{1}{n} \sum_{i=1}^n \left[\frac{C_{d,i} Y_i}{\hat{\pi}_{d,i}} - \frac{C_{d,i} - \hat{\pi}_{d,i}}{\hat{\pi}_{d,i}} \hat{m}_d(H_{1i}) \right].$$

- ▶ First term: IPW value estimator.
- ▶ Second term: augmentation with conditional mean zero if π_1 is correct.
- ▶ The augmentation often reduces instability from large weights.

Double Robustness

Define one-observation contribution

$$\phi(O; d, m, \pi) = m(H_1) + \frac{C_d}{\pi_d(H_1)} \{Y - m(H_1)\}.$$

Then $\mathbb{E}\{\phi(O; d, m, \pi)\} = V(d)$ if either:

- ▶ $m(H_1) = \mathbb{E}\{Y^*(d) \mid H_1\}$ is correct; or
- ▶ $\pi_d(H_1) = P(C_d = 1 \mid H_1)$ is correct.

If both are correct, AIPW is typically more efficient than IPW.

Double Robustness

AIPW is consistent if *either* nuisance model is correct.

	m correct	m wrong
π correct	consistent + efficient	consistent
π wrong	consistent	biased

From Evaluating a Rule to Learning One

For a *fixed* rule we treated $Y^*(d)$ as missing when $C_d = 0$, then estimated $V(d)$ by IPW or AIPW.

To **learn**, index rules by η and maximize the *same* estimator over the class $\mathcal{D}_\eta$:

$$\hat{\eta} \in \operatorname{argmax}_{\eta} \hat{V}_{\text{AIPW}}(d_\eta).$$

With per-action AIPW pseudo-outcomes

$$\hat{\psi}_a(O_i) = \hat{Q}_1(H_{1i}, a) + \frac{\mathbb{I}(A_{1i}=a)}{\hat{\pi}_a(H_{1i})} \{Y_i - \hat{Q}_1(H_{1i}, a)\} \text{ and contrast}$$
$$\hat{C}_i = \hat{\psi}_1(O_i) - \hat{\psi}_0(O_i):$$

$$\hat{V}_{\text{AIPW}}(d_\eta) = \frac{1}{n} \sum_i \left[d_1(H_{1i}; \eta) \hat{C}_i + \hat{\psi}_0(O_i) \right].$$

Only the first term depends on η : the learned rule picks action 1 where $\hat{C}_i > 0$.

Evaluation vs. Learning

Every value estimator does double duty — report a rule's value, or *become* the objective to maximize.

	Evaluate a fixed d	Learn d^{opt}
IPW	unbiased (known π), but inefficient — small weights inflate variance	maximize a <i>nonsmooth</i> $\hat{V}_{\text{IPW}}(d_\eta)$; brute-force grid search is tedious
AIPW	doubly robust and efficient — the preferred estimator	same idea, but augmentation makes it the <i>hardest</i> to optimize over η
Q-learning	plug-in $\frac{1}{n} \sum_i \max_a \hat{Q}_1(H_{1i}, a)$	easy: regress then argmax — but a misspecified $\hat{Q}_1$ gives a poor rule

Because the value is nonsmooth in the rule, specialized learners — OWL / weighted classification, policy search, and RL methods — replace grid search.

Casting Value Search into Classification

Value search can be rewritten as weighted classification.

Given the estimated contrast $\hat{C}_i$, the rule tries to match histories to the action with larger estimated value:

$$\text{label: } \mathbb{I}\{\hat{C}_i > 0\}, \quad \text{weight: } |\hat{C}_i|.$$

Larger estimated contrasts matter more for classification.

Outcome Weighted Learning

Outcome weighted learning (OWL) recasts value search as weighted classification — now weighting by the raw outcome, with the observed action as the label, rather than by $|\hat{C}_i|$.

With $Y \geq 0$, the IPW value counts rule-followers,

$$\hat{V}_{\text{IPW}}(d) = \frac{1}{n} \sum_{i=1}^n \frac{Y_i}{P(A_1=A_{1i} \mid H_1)} \mathbb{I}\{A_{1i} = d_1(H_{1i})\},$$

so maximizing value is minimizing outcome-weighted misclassification:

$$\min_{\eta_1} \frac{1}{n} \sum_{i=1}^n \frac{Y_i}{P(A_1=A_{1i} \mid H_1)} \mathbb{I}\{A_{1i} \neq d_1(H_{1i}; \eta_1)\}.$$

OWL with a Surrogate Loss

Code actions as $\{-1, 1\}$ and write

$$d_1(h_1; \eta_1) = \text{sign}\{f_1(h_1; \eta_1)\}.$$

The 0-1 loss can be replaced by hinge loss for better optimization (convexity):

$$\ell_{\text{hinge}}(u) = (1 - u)_+.$$

A typical OWL objective is

$$\frac{1}{n} \sum_{i=1}^n W_i \{1 - A_{1i} f_1(H_{1i}; \eta_1)\}_+ + \lambda_n \|f_1\|^2,$$

with $W_i = Y_i / P(A_1 = A_{1i} \mid H_1 = H_{1i})$.

- ▶ OWL assumes a randomized trial and $Y \geq 0$, and is sensitive to shifts in Y ; the doubly robust extension **EARL** removes these. ▶ EARL (appendix)

Indirect Route: Estimate Q_1 , Then Optimize

The regression route starts from the identified optimal rule:

$$d_1^{\text{opt}}(h_1) \in \operatorname{argmax}_{a_1 \in \mathcal{A}_1} Q_1(h_1, a_1).$$

Posit and fit a model $Q_1(h_1, a_1; \beta_1)$.

The estimated rule is

$$\hat{d}_{Q,1}^{\text{opt}}(h_1) \in \operatorname{argmax}_{a_1 \in \mathcal{A}_1} Q_1(h_1, a_1; \hat{\beta}_1).$$

“Q-learning” (Eventually)

A-Learning

For binary actions, it is easy to observe that the optimal rule depends only on the contrast

$$C_1(h_1) = Q_1(h_1, 1) - Q_1(h_1, 0).$$

A-learning models the contrast directly:

$$C_1(h_1; \psi_1), \quad \hat{d}_{A,1}^{\text{opt}}(h_1) = \mathbb{I}\{C_1(h_1; \hat{\psi}_1) > 0\}.$$

It avoids requiring a correct model for the full outcome regression $Q_1(h_1, a_1)$.

g-Estimation

- ▶ **What:** directly model the contrast/blip $C_1(h_1; \psi_1) = Q_1(h_1, 1) - Q_1(h_1, 0)$ (a structural nested mean model); the rule is $\mathbb{I}\{C_1(h_1; \psi_1) > 0\}$.
- ▶ **Why it works:** “blipping down” $Y - A_1 C_1(H_1; \psi_1)$ recovers the untreated outcome $Y^*(0)$, independent of A_1 given H_1 (no unmeasured confounding). At the true ψ_1 it is uncorrelated with the centered treatment $A_1 - \pi_1(H_1)$; a wrong ψ_1 leaves a treatment-correlated remainder — only the *propensity* is needed, the baseline is left free.

$$\sum_i \{A_{1i} - \pi_1(H_{1i})\} \lambda(H_{1i}) \{Y_i - A_{1i} C_1(H_{1i}; \psi_1)\} = 0.$$

Direct vs Indirect Learning

Direct value search

- ▶ Optimizes $\hat{V}(d_\eta)$ directly.
- ▶ Requires a specified $\mathcal{D}_\eta$.
- ▶ Natural for interpretable rules.
- ▶ IPW, AIPW, and OWL.

Indirect regression

- ▶ Estimates $Q_1(h_1, a_1)$ or $C_1(h_1)$.
- ▶ Optimizes over a_1 afterward.
- ▶ Rule class is induced by the model.
- ▶ Q, A-learning.

Practical Considerations

The performance target is the value of the learned rule:

$$V(\hat{d}^{\text{opt}}),$$

not just the numerical value of the optimized criterion:

$$\hat{V}(\hat{d}^{\text{opt}}).$$

- ▶ $\hat{V}(\hat{d}^{\text{opt}})$ can be optimistic after searching.
- ▶ IPW value search can be unstable when propensities are small.
- ▶ AIPW value search often improves stability and offers model-robustness.
- ▶ No method is uniformly best across data-generating mechanisms.

Practical Considerations

In practice:

- ▶ define a clinically interpretable rule class when direct search is used;
- ▶ fit more than one method when feasible;
- ▶ align rule classes across methods for a fair qualitative comparison;
- ▶ check whether conclusions agree in direction, not only in estimates;
- ▶ evaluate the chosen rule with sample splitting, cross-validation, or external data.

A stable *value* can still mask a **slippery rule**: where the contrast is near zero, tiny noise flips the decision and the value functional is nondifferentiable (*nonregularity*) — so naive standard errors and the ordinary bootstrap are invalid. Valid inference uses the *m-out-of-n* **bootstrap** (Chakraborty, Laber & Zhao, 2013) and empirical-process theory (Kosorok et al. 2008).

Section 2

Multi-Decision Setting

From One Decision to a Treatment Sequence

In the single-decision setting:

$$O_i = (H_{1i}, A_{1i}, Y_i).$$

In the finite-horizon setting, decisions occur at

$$k = 1, \dots, K.$$

A regime is now a sequence of rules:

$$d = (d_1, \dots, d_K).$$

Each rule uses the history available when that decision is made.

Finite Horizon Setup

Assume:

- ▶ K prespecified ordered decision points.
- ▶ $A_k \in \mathcal{A}_k$: action taken at decision k .
- ▶ X_1 : baseline information before decision 1.
- ▶ $X_k, k = 2, \dots, K$: new information between decisions $k - 1$ and k .
- ▶ Y : final outcome, coded so larger values are better.

For now: every individual is conceptualized as reaching all K decision points.

Notation

Let

$$\bar{X}_k = (X_1, \dots, X_k), \quad \bar{A}_k = (A_1, \dots, A_k).$$

The observed history available at decision k is

$$\begin{aligned} H_1 &= X_1, \\ H_k &= (\bar{X}_k, \bar{A}_{k-1}), \quad k = 2, \dots, K. \end{aligned}$$

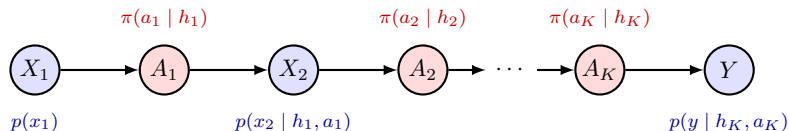
Equivalently,

$$H_k = (X_1, A_1, \dots, X_{k-1}, A_{k-1}, X_k).$$

A valid decision at time k can use H_k , but not future information.

The DGP as an Alternating Process

A trajectory is built one draw at a time, conditioning on the entire past h_k :



Nature draws baseline, covariate transitions, outcome; **Mechanism** draws the actions. In a SMART the red factors are *known by design*.

Observed Data as a Data-Generating Process

One observed trajectory is

$$O = (X_1, A_1, X_2, A_2, \dots, X_K, A_K, Y).$$

The temporal ordering is

$$X_1 \longrightarrow A_1 \longrightarrow X_2 \longrightarrow A_2 \longrightarrow \cdots \longrightarrow X_K \longrightarrow A_K \longrightarrow Y$$

- ▶ Observational data: actions chosen by clinicians, patients, systems.
- ▶ Sequentially randomized data: actions randomized at each decision.

A Multiple-Decision Regime

A finite-horizon regime is

$$d = (d_1, \dots, d_K),$$

where each rule maps current history to an action:

$$d_k : \mathcal{H}_k \rightarrow \mathcal{A}_k.$$

More generally, if only some actions are feasible at a history,

$$d_k(h_k) \in \Psi_k(h_k) \subseteq \mathcal{A}_k.$$

The feasible set $\Psi_k(h_k)$ lets the rule respect clinical eligibility and safety.

Following a Sequential Regime

If a subject follows d , the actions are generated recursively.

At decision 1:

$$A_1 = d_1(X_1).$$

At decision 2:

$$A_2 = d_2\{X_1, d_1(X_1), X_2\}.$$

In general:

$$A_k = d_k(H_k), \quad H_k = (\bar{X}_k, \bar{A}_{k-1}).$$

The history that the rule sees depends on previous rule-selected actions.

Potential Outcomes for Treatment Sequences

For a fixed treatment sequence $\bar{a}_K = (a_1, \dots, a_K)$:

- ▶ $X_k^*(\bar{a}_{k-1})$: information that would arise before decision k if earlier actions were $\bar{a}_{k-1}$.
- ▶ $Y^*(\bar{a}_K)$: final outcome if the full sequence $\bar{a}_K$ were assigned.

The full potential-outcome collection includes

$$\{X_1, X_2^*(a_1), \dots, X_K^*(\bar{a}_{K-1}), Y^*(\bar{a}_K)\}.$$

Later covariates are themselves potential outcomes of earlier treatment.

Potential Outcomes Under a Regime

Under a regime d , define the induced trajectory recursively:

$$\begin{aligned}A_k^d &= d_k(H_k^d), \\X_k^d &= X_k^*(\bar{A}_{k-1}^d), \quad k = 2, \dots, K, \\Y^d &= Y^*(\bar{A}_K^d) \equiv Y^*(d).\end{aligned}$$

Here $H_k^d = (\bar{X}_k^d, \bar{A}_{k-1}^d)$.

The value of the regime is

$$V(d) = \mathbb{E}(Y^d) = \mathbb{E}\{Y^*(d)\}.$$

An optimal regime in a class $\mathcal{D}$ satisfies

$$d^{\text{opt}} \in \operatorname{argmax}_{d \in \mathcal{D}} V(d).$$

Sequential SUTVA / Consistency

Sequential consistency says observed variables equal the corresponding potential variables under the actions actually received:

$$\begin{aligned}X_k &= X_k^*(\bar{A}_{k-1}), & k &= 2, \dots, K, \\Y &= Y^*(\bar{A}_K).\end{aligned}$$

- ▶ Treatment options are well-defined at each decision.
- ▶ No interference across individuals.
- ▶ Observed histories are compatible with the observed treatment path.

Sequential Randomization Assumption

Let W^* denote the full collection of potential histories and outcomes.

A strong form of the sequential randomization assumption is

$$W^* \perp\!\!\!\perp A_k \mid H_k, \quad k = 1, \dots, K.$$

Interpretation:

among subjects with the same observed history at decision k ,
treatment assignment at k is as good as random.

The required adjustment set changes with time.

Sequential Randomization

In multi-stage settings, later histories are often:

- ▶ predictors of future treatment;
- ▶ predictors of final outcome;
- ▶ affected by earlier treatment.

So X_2 can be both:

a confounder for the effect of A_2 and a mediator of the effect of A_1 .

Sequential methods do not adjust for everything all at once; they condition in time order and average over future histories under the regime.

- ▶ The SRA makes H_k a *sufficient adjustment set* at stage k : condition on the confounders it contains, and treatment is as good as random — but you cannot adjust for them all at once. ▶ Why naive adjustment fails

Sequential Positivity

For each decision k , histories with support must contain support for all feasible actions:

$$P(A_k = a_k \mid H_k = h_k) > 0, \quad a_k \in \Psi_k(h_k), \quad k = 1, \dots, K.$$

To evaluate a fixed regime d :

$$P\{A_k = d_k(H_k) \mid H_k\} > 0 \quad \text{along the regime path.}$$

Sequential positivity is a product-over-time problem.

Single $\rightarrow$ Multiple

As before, our assumptions allow us to write the distribution of potential outcomes under a fixed regime using only the observed sequences of data.

	Single decision	K decisions
Value	$\mathbb{E}\{Y^*(d)\}$	same, for $d = (d_1, \dots, d_K)$
IPW	weight $1/\pi_{d,1}$	product $1/\prod_k \pi_{d,k}$
AIPW	one residual correction	sequential correction
Regression	Q_1 , then argmax	backward induction $Q_K \rightarrow Q_1$
A-learning	blip $C_1(h_1)$	stage blips $C_k(h_k)$
Classification	OWL	backward OWL

Regression Modeling

value of d_1 depends on the *entire* future



solve backward: $Q_K \rightarrow \dots \rightarrow Q_1$

(implement forward $\rightarrow$ solve backward)

- ▶ We implement forward in time, but cannot *optimize* forward: the best d_1 depends on the whole future strategy.
- ▶ $Q_k(h_k, a_k)$: the expected value of choosing a_k now, then acting optimally afterward.
- ▶ V_{k+1} : the best achievable future from the next history onward — each Q_k needs it.
- ▶ The last decision has no future to weigh, so solve it first, then work back.

G-Computation: Two Decisions

For $K = 2$, a fixed regime $d = (d_1, d_2)$ has value

$$V(d) = \mathbb{E} \left[\mathbb{E} \{ Y \mid X_1, A_1 = d_1(X_1), X_2, A_2 = d_2(X_1, d_1(X_1), X_2) \} \right],$$

where the outer expectation averages over the distribution of

$$X_1 \quad \text{and} \quad X_2 \mid X_1, A_1 = d_1(X_1).$$

The later history is drawn under the earlier rule-selected action.

G-Computation: General Form

Write $\bar{a}_k^d(\bar{x}_k)$ for the treatment history assigned by d along covariate history $\bar{x}_k$.

G-computation uses

$$p_{Y|\bar{X}_K, \bar{A}_K} \{y \mid \bar{x}_K, \bar{a}_K^d(\bar{x}_K)\}$$

and the covariate evolution densities

$$p_{X_k|\bar{X}_{k-1}, \bar{A}_{k-1}} \{x_k \mid \bar{x}_{k-1}, \bar{a}_{k-1}^d(\bar{x}_{k-1})\}.$$

In words:

- ▶ set treatment to the rule at each time;
- ▶ model how future histories evolve under those actions;
- ▶ average predicted outcomes over those future histories.

G-Computation by Iterated Regressions

A practical regression version works backward.

1. Regress Y on (H_K, A_K) .
2. Predict at $A_K = d_K(H_K)$ to create a pseudo-outcome.
3. Regress that pseudo-outcome on (H_{K-1}, A_{K-1}) .
4. Predict at $A_{K-1} = d_{K-1}(H_{K-1})$.
5. Continue backward to decision 1.

Finally average the decision-1 predictions over the empirical distribution of H_1 .

IPW Evaluation for a Fixed Regime

Define the indicator that a subject followed the regime through all K decisions:

$$C_d = \prod_{k=1}^K \mathbb{I}\{A_k = d_k(H_k)\}.$$

Let

$$\pi_{d,k}(H_k) = P\{A_k = d_k(H_k) \mid H_k\}.$$

Under the identifying assumptions,

$$V(d) = \mathbb{E} \left[\frac{C_d Y}{\prod_{k=1}^K \pi_{d,k}(H_k)} \right].$$

IPW Estimator

The simple multi-stage IPW estimator is

$$\hat{V}_{\text{IPW}}(d) = \frac{1}{n} \sum_{i=1}^n \frac{C_{d,i} Y_i}{\prod_{k=1}^K \hat{\pi}_{d,k}(H_{ki})}.$$

A normalized version is

$$\hat{V}_{\text{IPW}*}(d) = \frac{\sum_{i=1}^n C_{d,i} Y_i / \prod_{k=1}^K \hat{\pi}_{d,k}(H_{ki})}{\sum_{i=1}^n C_{d,i} / \prod_{k=1}^K \hat{\pi}_{d,k}(H_{ki})}.$$

In a sequentially randomized study, the $\pi_{d,k}$ are known by design.

Why Multi-Stage IPW Gets Fragile

The denominator is a product:

$$\prod_{k=1}^K \hat{\pi}_{d,k}(H_k).$$

Similar to before:

- ▶ Products of moderate probabilities can become small.
- ▶ Only subjects with $C_d = 1$ contribute outcome information.
- ▶ The number following a full rule can shrink quickly with K .
- ▶ Large weights make estimates noisy and sensitive to a few trajectories.

AIPW Evaluation: Partial Information

For a fixed regime, define

$$C_{d,k} = \prod_{j=1}^k \mathbb{I}\{A_j = d_j(H_j)\}, \quad \bar{\pi}_{d,k} = \prod_{j=1}^k \pi_{d,j}(H_j),$$

with $C_{d,0} = \bar{\pi}_{d,0} = 1$.

AIPW can be written as

$$\hat{V}_{\text{AIPW}}(d) = \frac{1}{n} \sum_i \left[\frac{C_{d,K,i} Y_i}{\hat{\pi}_{d,K,i}} + \sum_{k=1}^K \left\{ \frac{C_{d,k-1,i}}{\hat{\pi}_{d,k-1,i}} - \frac{C_{d,k,i}}{\hat{\pi}_{d,k,i}} \right\} \hat{L}_k(H_{ki}) \right].$$

where $\hat{L}_k(H_{ki})$ is our outcome regression model

AIPW Evaluation: Interpretation

In the multi-stage setting, AIPW treats rule-following like a monotone missing-data problem.

- ▶ If a subject follows d through all K decisions, $Y^*(d)$ is observed.
- ▶ If the subject stops following d at decision k , their later observed Y is not under d .
- ▶ Their history through k still helps predict what would happen under d .

With suitable nuisance models, AIPW can be more stable than simple IPW and can be doubly robust.

Backward Outcome Weighted Learning (BOWL)

OWL doesn't generalize directly to K stages: the best rule at stage t depends on *future* decisions. **BOWL** (Zhao et al., 2015) solves it *backward* as a sequence of weighted classifications.

Stage K : ordinary OWL. **Stage $t = K-1, \dots, 1$:** among subjects who *followed* the estimated future rules, learn $d_t = \text{sign}(f_t)$ by minimizing

$$\frac{1}{n} \sum_i \underbrace{\frac{(\sum_{j \geq t} R_{ij}) \prod_{j > t} \mathbb{I}\{A_{ij} = \hat{d}_j(H_{ij})\}}{\prod_{j \geq t} \pi_j(A_{ij}, H_{ij})}}_{\text{weight } W_{it}} \{1 - A_{it} f_t(H_{it})\}_+ + \lambda \|f_t\|^2.$$

- ▶ Needs only the **propensities**.
- ▶ Not doubly robust

Backward Induction

Take $K = 2$ A patient has *reached* Decision 2, so their history

$H_2 = (X_1, A_1, X_2)$ is already determined.

With H_2 fixed, Decision 2 is just a *single* decision with H_2 as “baseline”:

$$Q_2(h_2, a_2) = \mathbb{E}(Y \mid H_2 = h_2, A_2 = a_2),$$

$$d_2^{\text{opt}}(h_2) = \operatorname{argmax}_{a_2 \in \mathcal{A}_2} Q_2(h_2, a_2),$$

$$V_2(h_2) = \max_{a_2 \in \mathcal{A}_2} Q_2(h_2, a_2).$$

$V_2(h_2)$ is the best expected outcome for someone with history h_2 who then follows d_2^{opt} .

Backward Induction

A patient with baseline H_1 takes A_1 , then sees X_2 , then follows d_2^{opt} — so their eventual expected outcome is $V_2(H_2)$, not Y directly.

The right “outcome” to optimize at Decision 1 is therefore V_2 :

$$Q_1(h_1, a_1) = \mathbb{E}\{V_2(H_2) \mid H_1 = h_1, A_1 = a_1\},$$

$$d_1^{\text{opt}}(h_1) = \operatorname{argmax}_{a_1 \in \mathcal{A}_1} Q_1(h_1, a_1), \quad V_1(h_1) = \max_{a_1 \in \mathcal{A}_1} Q_1(h_1, a_1).$$

Unfolded, this is a **maximization of the expected maximum**:

$$V_1(x_1) = \max_{a_1} \mathbb{E} \left[\max_{a_2} \mathbb{E}(Y \mid X_1=x_1, X_2, A_2=a_2) \mid X_1=x_1, A_1=a_1 \right].$$

Value of the optimal regime: $\mathbb{E}\{Y^*(d^{\text{opt}})\} = \mathbb{E}\{V_1(H_1)\}.$

Backward Induction: Final Decision

At the final decision, define

$$Q_K(h_K, a_K) = \mathbb{E}(Y \mid H_K = h_K, A_K = a_K).$$

The optimal final rule is

$$d_K^{\text{opt}}(h_K) \in \operatorname{argmax}_{a_K \in \Psi_K(h_K)} Q_K(h_K, a_K).$$

The corresponding value function is

$$V_K(h_K) = \max_{a_K \in \Psi_K(h_K)} Q_K(h_K, a_K).$$

At the last decision, the problem looks like a single-decision problem with baseline H_K .

Backward Induction: Earlier Decisions

For $k = K - 1, \dots, 1$, define

$$Q_k(h_k, a_k) = \mathbb{E}\{V_{k+1}(H_{k+1}) \mid H_k = h_k, A_k = a_k\}.$$

Then

$$d_k^{\text{opt}}(h_k) \in \operatorname{argmax}_{a_k \in \Psi_k(h_k)} Q_k(h_k, a_k),$$

$$V_k(h_k) = \max_{a_k \in \Psi_k(h_k)} Q_k(h_k, a_k).$$

The value of action a_k is judged assuming optimal future decisions afterward.

Q-Learning: Multi-Stage Algorithm

Q-learning estimates the Q_k functions backward.

1. Fit a model $Q_K(h_K, a_K; \beta_K)$ for $\mathbb{E}(Y \mid H_K, A_K)$.
2. Compute

$$\tilde{V}_{K,i} = \max_{a_K \in \Psi_K(H_{Ki})} Q_K(H_{Ki}, a_K; \hat{\beta}_K).$$

3. For $k = K - 1, \dots, 1$, fit $Q_k(h_k, a_k; \beta_k)$ using $\tilde{V}_{k+1,i}$ as the outcome.
4. Estimate each rule by maximizing the fitted Q_k over feasible actions.

Q-Learning Estimating Equations

At the final decision:

$$\sum_{i=1}^n \frac{\partial Q_K(H_{Ki}, A_{Ki}; \beta_K)}{\partial \beta_K} \{Y_i - Q_K(H_{Ki}, A_{Ki}; \beta_K)\} = 0.$$

For earlier decisions, replace Y_i by the next pseudo-outcome:

$$\sum_{i=1}^n \frac{\partial Q_k(H_{ki}, A_{ki}; \beta_k)}{\partial \beta_k} \{\tilde{V}_{k+1,i} - Q_k(H_{ki}, A_{ki}; \beta_k)\} = 0.$$

Two-Stage Linear Q-Learning

For binary actions and two stages, a common model is

$$Q_2(h_2, a_2; \beta_2) = \beta_{21}^T \tilde{h}_2 + a_2 \beta_{22}^T \tilde{h}_2,$$

$$Q_1(h_1, a_1; \beta_1) = \beta_{11}^T \tilde{h}_1 + a_1 \beta_{12}^T \tilde{h}_1.$$

Then the estimated rules are

$$\hat{d}_{Q,2}^{\text{opt}}(h_2) = \mathbb{I}\{\hat{\beta}_{22}^T \tilde{h}_2 > 0\},$$

$$\hat{d}_{Q,1}^{\text{opt}}(h_1) = \mathbb{I}\{\hat{\beta}_{12}^T \tilde{h}_1 > 0\}.$$

A-Learning: Multi-Stage Contrast Models

For binary actions, define the stage- k contrast

$$C_k(h_k) = Q_k(h_k, 1) - Q_k(h_k, 0).$$

The optimal rule depends on the sign:

$$d_k^{\text{opt}}(h_k) = \mathbb{I}\{C_k(h_k) > 0\}.$$

A-learning posits contrast models

$$C_k(h_k; \psi_k), \quad k = 1, \dots, K,$$

rather than full models for every $Q_k(h_k, a_k)$.

A-Learning Pseudo-Outcomes

For binary actions with option 0 as reference, a backward update has the form

$$\tilde{V}_{k,i} = \tilde{V}_{k+1,i} + \max\{0, C_k(H_{ki}; \hat{\psi}_k)\} - A_{ki}C_k(H_{ki}; \hat{\psi}_k),$$

with $\tilde{V}_{K+1,i} = Y_i$.

Interpretation:

- ▶ remove the observed treatment blip;
- ▶ add back the optimal blip;
- ▶ move one step earlier.

Direct Value Search for Multiple Decisions

Choose a restricted class

$$\mathcal{D}_\eta = \{d_\eta = (d_1(\cdot; \eta_1), \dots, d_K(\cdot; \eta_K))\}.$$

The target is

$$\eta^{\text{opt}} \in \operatorname{argmax}_\eta V(d_\eta).$$

Estimate by maximizing a value estimator:

$$\hat{\eta}^{\text{opt}} \in \operatorname{argmax}_\eta \hat{V}(d_\eta).$$

$\hat{V}$ may be IPW, AIPW, or another fixed-regime value estimator.

Direct Search: Global vs Backward

Global value search:

$$\hat{\eta}^{\text{opt}} = \operatorname{argmax}_{\eta_1, \dots, \eta_K} \hat{V}(d_{\eta}).$$

Backward iterative search:

- ▶ optimize the final rule first;
- ▶ hold that estimated future rule fixed;
- ▶ move backward one decision at a time.

Backward search lowers the dimension of each optimization but may target the global restricted optimum only under additional conditions.

Evaluation vs. Learning

Evaluation

- ▶ Fixed d .
- ▶ Estimate $V(d)$.
- ▶ G-computation, IPW, AIPW.
- ▶ Diagnostics: support, weights, model fit.

Learning

- ▶ Unknown d .
- ▶ Estimate high-value rules.
- ▶ Direct: value search, OWL-type methods.
- ▶ Indirect: Q-learning, A-learning.

Practical Considerations: Assumptions

- ▶ H_k must contain the information used to choose treatment at decision k .
- ▶ This requirement is harder in observational sequential data than in point-exposure data.
- ▶ Sequential positivity requires support for each feasible option within relevant histories.
- ▶ Feasible sets $\Psi_k(h_k)$ should be scientifically justified and as simple as the context allows.
- ▶ In sequentially randomized designs, SRA and positivity are partly design targets.

Practical Considerations: Estimation

- ▶ G-computation avoids treatment models but requires models for how histories evolve.
- ▶ Simple IPW uses outcome information only from trajectories consistent with the rule.
- ▶ Multi-stage IPW is realistic mainly when K and the number of options are modest.
- ▶ Q-learning is intuitive, but earlier regressions use pseudo-outcomes so misspecification of an outcome model can compound.
- ▶ Direct search faces nonsmooth optimization, now over parameters for multiple rules.

Methods and Literature

Estimating the regime

- ▶ *Regression (indirect)*: Q- / IQ-learning (Murphy 2003; Laber et al. 2014a); flexible fitted-Q (Ernst et al. 2005; Moodie et al. 2014).
- ▶ *Contrast*: A-learning / g-estimation of structural nested mean models (Murphy et al. 2003, Blatt et al. 2004, Robins 2004; Moodie et al. 2007).
- ▶ *Direct value / policy search*: AIPW value & classification search (Zhang et al. 2012, 2013); OWL / BOWL / SOWL / EARL (Zhao et al. 2012, 2015, 2019).
- ▶ *Interpretable*: decision lists (Zhang et al. 2015), trees (Laber & Zhao 2015).

Inference and more

- ▶ *Inference / nonregularity*: m -out-of- n bootstrap (Chakraborty et al. 2013), adaptive CIs (Laber et al. 2014b), penalized Q (Song et al. 2015).
- ▶ Q-learning vs. A-learning (Chakraborty et al. 2010; Schulte et al. 2014).
- ▶ *Special settings*: survival (Goldberg & Kosorok 2012), resource / budget constraints (Luedtke & van der Laan 2016), continuous doses (Chen et al. 2016), variable selection (Gunter et al. 2011).
- ▶ Bayesian: learning the value-function surface with Gaussian processes (Freeman et al. 2024), Bayesian OWL (Yazzourh et al. 2024).

SMART_s

A **SMART** (sequential multiple assignment randomized trial) randomizes treatment at *each* decision point — the multi-stage analogue of a randomized trial .

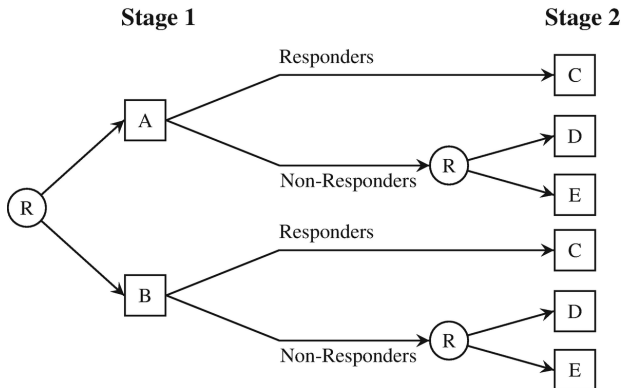


Figure: Almirall et al. 2021

Why a SMART Makes Estimation Easier

In the DGP factorization, a SMART makes the **treatment mechanism** known:

- ▶ **SRA holds by design.** Randomizing given history $\Rightarrow$ assignment $\perp\!\!\!\perp$ potential outcomes: no unmeasured confounding to argue (“gold standard”).
- ▶ **Positivity holds by design.** Each feasible option is assigned with known positive probability.
- ▶ **Propensities are known.** $p(a_k | h_k)$ is set by the design, not estimated:
 - ▶ IPW/AIPW weights are known — *no propensity misspecification*;
 - ▶ plug in the known π_k .

The only remaining work is modeling **nature** (Q -functions / transitions).

Section 3

Infinite Time Horizon

Indefinite Horizons Change the Target

We will use **indefinite horizon** and **infinite horizon** interchangeably here.

Finite horizon

$$d = (d_1, \dots, d_K)$$

- ▶ K is fixed in advance.
- ▶ Rules may depend on decision time.
- ▶ Backward induction starts at K .

Indefinite horizon

$$\pi : \mathcal{S} \rightarrow \mathcal{A}$$

- ▶ T is unknown, subject-specific, or infinite.
- ▶ The same rule can be applied repeatedly.
- ▶ There is no final stage to start from.

A Chronic-Disease Example: Type 2 Diabetes

Managing a chronic condition is naturally *indefinite-horizon*: therapy is revisited at every clinic visit, for years, with no prespecified final decision.

- ▶ **State** S_t : current A1c, comorbidities, current medication.
- ▶ **Action** A_t : maintain, adjust, or intensify glucose-lowering therapy.
- ▶ **Reward** R_{t+1} : better A1c control, coded so larger is better.

Goal: one **stationary** rule $\pi(s)$, reused at every visit and applicable *beyond* the observed follow-up.

Observed Data Still End

A typical observed trajectory is finite:

$$O_i = (S_{0i}, A_{0i}, S_{1i}, A_{1i}, \dots, S_{\tilde{T}_i, i}).$$

The scientific target may still be a rule that can be applied beyond the observed horizon.

We learn from observed transitions:

$$(S_{ti}, A_{ti}, R_{t+1,i}, S_{t+1,i}), \quad t = 0, \dots, \tilde{T}_i - 1.$$

The data are finite; the proposed regime is not indexed by a terminal decision time.

States, Actions, and Rewards

- ▶ $S_t \in \mathcal{S}^\dagger$: state summary available before action at time t .
- ▶ $A_t \in \mathcal{A}_{S_t}$: feasible action in the current state.
- ▶ S_{t+1} : next state after action A_t .
- ▶ $R_{t+1} = r(S_t, A_t, S_{t+1})$: immediate reward, coded so larger is better.

For terminal events, let $\mathcal{S}^\dagger = \mathcal{S} \cup \{c\}$, where c is absorbing:

$$S_t = c \implies S_{t+1} = c, \quad A_t = u, \quad R_{t+1} = 0.$$

Reinforcement Learning $\leftrightarrow$ DTRs

Concept	RL term	DTR term	Symbol
Decision	action	treatment	A_t, a
Context	state	covariate / history	S_t, s
Utility	reward	outcome	R_t, Y
Unit	trajectory	patient	i
Time index	time step	stage / visit	t
Strategy	policy	(dynamic) regime	π, d
Support	exploration / coverage	positivity	—

Stationarity

A deterministic policy is

$$\pi(s) \in \mathcal{A}_s.$$

A stochastic policy assigns probabilities:

$$\pi(a \mid s), \quad a \in \mathcal{A}_s.$$

Stationary does **not** mean the patient is unchanged. It means the same mapping from current state to action is used at every decision time.

The state must carry enough history for that mapping to be meaningful.

Terminology

How data arrive

- ▶ **Offline/batch**: learn from already-collected data.
- ▶ **Online**: update the policy while data accrue.

Which policy generated actions

- ▶ **Behavior policy**: observed action mechanism.
- ▶ **Target policy**: policy to evaluate or learn.

Policy alignment

- ▶ **On-policy**: behavior and target policies match.
- ▶ **Off-policy**: target differs from behavior.

Action assignment

- ▶ **Exploration**: assign actions to learn.
- ▶ **Exploitation**: assign the current best action.

Value Over an Unbounded Sequence

For a policy π , define

$$V^{\pi}(s) = \mathbb{E}^{\pi} \left[\sum_{k=1}^{\infty} \gamma^{k-1} R_{t+k} \mid S_t = s \right],$$
$$Q^{\pi}(s, a) = \mathbb{E}^{\pi} \left[\sum_{k=1}^{\infty} \gamma^{k-1} R_{t+k} \mid S_t = s, A_t = a \right].$$

The discount factor $0 \leq \gamma < 1$ controls how strongly future rewards matter.

A population target can average over a reference distribution R :

$$V_R(\pi) = \int V^{\pi}(s) dR(s).$$

Assumptions for the Indefinite Horizon

The three causal assumptions carry over (rewards in place of a final Y):

- ▶ **Consistency**: observed rewards equal the potential rewards under the actions taken.
- ▶ **Sequential randomization**: no unmeasured confounders given history.
- ▶ **Positivity / exploration**: $P(A_t = a \mid \bar{S}_t, \bar{A}_{t-1}) > 0$ for every feasible a .

New: a time-homogeneous Markov decision process (MDP)

— a state S_t summarizing history with

$$\text{Markov: } S_{t+1} \perp\!\!\!\perp (\bar{S}_{t-1}, \bar{A}_{t-1}) \mid S_t, A_t,$$

$$\text{time-homogeneous: } P(S_{t+1}=s' \mid S_t=s, A_t=a) = p(s' \mid s, a),$$

reward $R_{t+1} = r(S_t, A_t, S_{t+1})$, and discount $0 \leq \gamma < 1$.

Why: there is no terminal stage to induct from, and an ever-growing history cannot index a reusable rule — a Markov state lets one *stationary* policy be applied (and extrapolated) beyond the observed data.

The Bellman Equations

Policy evaluation. The value of a fixed policy π solves a fixed-point recursion:

$$\begin{aligned}Q^\pi(s, a) &= \mathbb{E}[R_{t+1} + \gamma Q^\pi\{S_{t+1}, \pi(S_{t+1})\} \mid S_t=s, A_t=a], \\V^\pi(s) &= \sum_a \pi(a \mid s) Q^\pi(s, a).\end{aligned}$$

Optimality. Q^* is the unique fixed point of the Bellman *optimality* equation:

$$\begin{aligned}Q^*(s, a) &= \mathbb{E}\left[R_{t+1} + \gamma \max_{a' \in \mathcal{A}_{S_{t+1}}} Q^*(S_{t+1}, a') \mid S_t=s, A_t=a\right], \\ \pi^*(s) &\in \operatorname{argmax}_a Q^*(s, a).\end{aligned}$$

Model-Based Evaluation: Estimate the Dynamics

Estimate the two pieces of the MDP from observed transitions $(S_t, A_t, R_{t+1}, S_{t+1})$:

- ▶ **Transition model** $\hat{p}(s' | s, a)$: model the next state given (S_t, A_t) — a parametric kernel, a conditional density, or a fitted regression $S_{t+1} = f(S_t, A_t) + \varepsilon$.
- ▶ **Reward model**: $r(s, a, s')$ is often a *known* utility (e.g. a function of A1c); if the reward is stochastic, estimate $\hat{r}(s, a) = \mathbb{E}(R_{t+1} | S_t=s, A_t=a)$ by regression.

Plug the fitted models into the Bellman equation and solve for V^π by value iteration, or *simulate* trajectories under π (Monte-Carlo rollouts):

$$\hat{V}^\pi(s) = \mathbb{E}_{\hat{p}} \left[R_{t+1} + \gamma \hat{V}^\pi(S_{t+1}) \mid S_t=s, A_t=\pi(s) \right].$$

Analogue of g-computation.

MDP Transition Probabilities

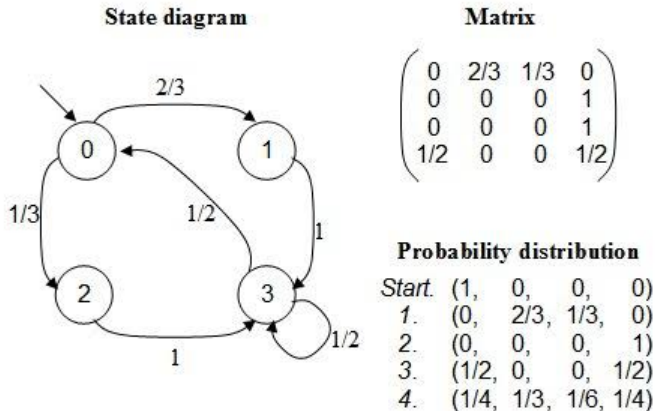


Figure: Phil Hooker, University of Edinburgh

Direct Search Becomes Policy Search

Choose an interpretable policy class

$$\Pi_{\eta} = \{\pi_{\eta} : \eta \in \mathbb{R}^p\}.$$

Then search for high-value policies:

$$\hat{\eta} \in \operatorname{argmax}_{\eta} \hat{V}_R(\pi_{\eta}).$$

Infinite-horizon policy-search estimators can use:

- ▶ IPW or AIPW value estimates;
- ▶ Value function estimating equations;
- ▶ Q-function estimates to stabilize the objective.

Practical Considerations

Make the estimand credible

- ▶ **State construction:** S_t must be rich enough for the future to be approximately Markov and time-homogeneous.
- ▶ **Reward and discount:** $r(S_t, A_t, S_{t+1})$ and γ encode the clinical tradeoff between short- and long-run outcomes.
- ▶ **Coverage:** the behavior policy must try every action the target policy might recommend in relevant states.

Make the estimator deployable

- ▶ **Offline data:** off-policy evaluation is fragile; long products of importance ratios can have very high variance.
- ▶ **Approximation:** features / policy classes should be interpretable and checked with Bellman or temporal-difference residuals.
- ▶ **Online data:** exploration is necessary, but must be randomized, monitored, and ethically constrained.

Methods and Literature

A map of RL methods for indefinite / infinite horizons:

Estimating the policy

- ▶ *Model-based*: fit transition & reward, then plan / simulate; hybrids (MAGIC, Thomas & Brunskill 2016).
- ▶ *Model-free value / Q*: fitted Q-iteration (Ernst et al. 2005), batch infinite-horizon Q-learning / gradient-TD (Ertefaie & Strawderman 2018; Maei et al. 2010).
- ▶ *Policy search / actor-critic*: V-learning (Luckett et al. 2019) [▶ details](#); policy gradients.

RL-specific

- ▶ *Off-policy evaluation*: importance sampling; doubly robust OPE (Jiang & Li 2015).
- ▶ *Exploration / online*: contextual bandits (Tewari & Murphy 2017), Thompson sampling.
- ▶ *Deep RL*: deep Q-networks (Mnih et al. 2015), TRPO / PPO (Schulman et al. 2015, 2017).
- ▶ *Applications & design*: mHealth micro-randomized trials (Klasnja et al. 2015, Susan Murphy), JITAs (Nahum-Shani et al. 2018), spatiotemporal allocation (Laber et al. 2018).

Review paper: Clifton & Laber 2020

Appendix

Supplementary derivations and details

Under the identifying assumptions,

$$V(d) = \mathbb{E} \left\{ \frac{C_d Y}{\pi_{d,1}(H_1)} \right\}.$$

$$\begin{aligned} \mathbb{E} \left\{ \frac{C_d Y}{\pi_{d,1}(H_1)} \right\} &= \mathbb{E} \left\{ \frac{C_d Y^*(d)}{\pi_{d,1}(H_1)} \right\} \\ &= \mathbb{E} \left[Y^*(d) \frac{\mathbb{E}(C_d \mid H_1, Y^*(0), Y^*(1))}{\pi_{d,1}(H_1)} \right] \\ &= \mathbb{E}\{Y^*(d)\}. \end{aligned}$$

Beyond OWL: EARL (Doubly Robust OWL) (appendix)

OWL's original form needs a randomized trial and $Y \geq 0$, and is sensitive to shifts in Y . **EARL** (Zhao, Laber, Ning, Saha & Sands, 2019) removes all three.

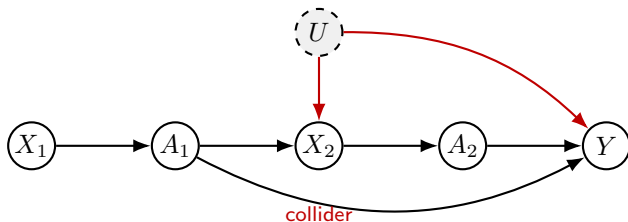
Replace the raw outcome weight by an **AIPW pseudo-outcome** for each action a :

$$\widehat{W}_a = \frac{Y \mathbb{I}(A_1=a)}{\widehat{\pi}(a \mid H_1)} - \frac{\mathbb{I}(A_1=a) - \widehat{\pi}(a \mid H_1)}{\widehat{\pi}(a \mid H_1)} \widehat{Q}_1(H_1, a),$$

then learn $\text{sign}(f)$ with weights $|\widehat{W}_a|$ and labels $\text{sign}(\widehat{W}_a)$.

- ▶ **Doubly robust**: consistent if *either* $\widehat{\pi}$ or $\widehat{Q}_1$ is correct (rates improve if both; hinge fastest).
- ▶ Works on **observational** data; Y need not be bounded or positive.
- ▶ **Relaxation**: any convex surrogate (hinge, logistic, exponential), with cross-fitting.

Why Naive Adjustment Fails: X_2 Is a Collider (appendix)



- ▶ X_2 is a **confounder for** A_2 ($X_2 \rightarrow A_2$, X_2 linked to Y via U) — stage 2 *must* condition on it.
- ▶ But X_2 is a **collider for** A_1 : $A_1 \rightarrow X_2 \leftarrow U \rightarrow Y$. Adjusting for X_2 to judge A_1 opens a spurious path (and blocks A_1 's effect *through* X_2).
- ▶ So no single regression is right for both stages — **condition in time order**: use X_2 at stage 2, never “adjust it away” for stage 1.

V-learning: Policy Search for Indefinite Horizons (appendix)

A policy-search method built directly on the Bellman equation for V^π (Lockett et al., 2019).

For a class of policies π , model the V-function $V(s; \eta_\pi)$ and solve the importance-weighted estimating equation

$$\mathbb{E} \left[\sum_t \rho_t(\pi; b) \{R_{t+1} + \gamma V(S_{t+1}; \eta_\pi) - V(S_t; \eta_\pi)\} \nabla_{\eta_\pi} V(S_t; \eta_\pi) \right] = 0,$$

with $\rho_t(\pi; b) = \pi(A_t | S_t) / b(A_t | S_t)$. Then choose the policy with the largest estimated value:

$$\hat{\pi} \in \operatorname{argmax}_{\pi} \int V(s; \eta_\pi) dR(s).$$

Off-policy, works from a single observational stream, and needs only a V-model rather than the full transition dynamics.

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